Veche: The Assembly of Humans' Recommenders

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Abstract

Some of the leading social medias' recommendation AIs have recently been argued and legally judged to be part of an "addictive design", both in the US and in the EU. Unfortunately, given a lack of alternatives, the only recommended alternative solution is too often the reverse chronological feed. Yet, while its transparency and simplicity indeed make it clearly less of a "dark pattern", this feed has plenty of undesirable properties. Namely, it provides a poor user experience, favors hyperactive users, and incentives massive content creation. In this paper, we introduce Veche, an alternative flexible follow-based recommendation algorithm, whose principles are well-grounded in social choice theory. Namely, Veche views a follow as an allocation of a unit voting right of influence. Followed users are asked what they want to recommend, and Veche constructs a feed out of their recommendation submissions. Veche thereby provides a rigorous traceability of influence. We also present various generalizations that respond to practical needs, and implementations that we deployed on Tournesol and on BlueSky. Overall, we make the case that Veche should replace the reverse chronological feed as the default follow-based feed.

1 Introduction

Social media have been recently sentenced for their endangerement of the mental health of young people. In the Netherlands, this has led to an imposition on Meta to reintroduce a reverse chronological feed [Reu26]. This feed is also widely used by alternative social media, like Mastodon, PeerTube and BlueSky. However, even though it is extremely interpretable, the reverse chronology recommendation algorithm fails to provide a satisfactory user experience. Additionally, it favors hyperactive publishers, and incentivizes fast and massive content creation. More generally, it does not fully address the demands for influence traceability.

In this paper, we propose Veche[1], an alternative, modular and flexible following-based recommendation algorithm grounded in social choice theory principles. Veche implements the following key principle.

A follow is the allocation of a unit voting right of influence.

More specifically, Veche considers that, when Alice follows some user u, the user u is given a unit voting right in a social choice mechanism that will then (stochastically) construct Alice's recommendation feed. Put differently, the user u then gets to submit a ballot b_u. A voting system will then aggregate all followed users' ballots, and return a feed based on these ballots, with the guarantee that each user u's influence is bounded by their unit voting right, no matter what ballot b_u is submitted.

Given that a ballot b_u must be be submitted upon any of Alice's query for a feed, and given that she expects to receive a feed within a fraction of a second, the construction of b_u will have to be automated,

[1]Veche was a popular assembly among some Slavic peoples during the Middle Ages.
The name is not set in stone yet, but I would strongly want it to promote democracy. Other candidates include Túath, Ecclesia or Landsgemeinde. Ideally, I'd love something close in spirit to the algorithm (i.e. selecting who you listen and gathering them to vote on what you should know). Feel free to check `https://en.wikipedia.org/wiki/History_of_democracy` or other sources to come up with other proposals.

based on user u's past activities (posts, reposts, reports...). The ballot-construction mechanism may called an *algorithmic representative*, following the principles put forth by [LKK+19, NGA+18, FBS+20]. While any user u could construct any algorithm to represent them by submitting ballots b_u, most users will require a default construction of an algorithmic representative. The paper also proposes a default algorithmic representative for all users.

More concretely, our contributions are the following.

- We introduce VECHE, a highly flexible social choice mechanism, through which followed users can vote for the content of Alice's feed, and we provide mathematical guarantees on the maximal impact of a followed user.
- We show how to leverage VECHE's flexibility and algorithmic representatives to propose a general and practical deployment that responds to user needs and constraints, such as minimal user experience modifications, simple time decay management with desirable mathematical properties, tunable followed users' volumes, including preferred semantic biases, encouraging reposts and reports over posts, and feed diversification using Tillé's pivot algorithm **[CITE]**.
- We provide mathematical proofs that VECHE and some of its proposed instantiation details are well-behaved, thereby providing interpretability and trustworthiness.
- We provide a free and open-source implementation of VECHE, as part of the Solidago library [HBC+22]. We also deploy VECHE on Tournesol [HBF+25] and BlueSky [KFG+24]. We also provide the materials to effortlessly adapt and redeploy our algorithms through the AT protocol [KFG+24].

Overall, we believe that the desirable properties of our algorithm, especially in terms of the traceability of influence, make a compelling case to make it, perhaps through legal enforcement, the default recommendation algorithm for all social media, instead of the reverse chronological feed.

The rest of the paper is organized as follows. First, Section 2 provides some context and a review of related literature. Next, for mostly pedagogical reasons, Section 3 opens with a simple instantiation of VECHE, and provides proofs of its main guarantees. Section 4 then shows generalization of the first instantiation, to highlight the flexibility of VECHE, and to show how to respond to various needs in practical deployments. Section 5 evaluates VECHE and other recommendation algorithms along Tournesol's four digital democracy pillars. Section 6 reports our empirical analysis. Section 7 concludes.

2 Context

The classical approach to the design of recommendation algorithms strongly relies on prediciting users' preference signals (e.g., likes or ratings) [BOHG13, ZB23, PPCC26]. Such a prediction method can leverage user and/or item similarities [SMH24], (nonnegative) matrix factorization [ABR+25], deep learning algorithms [CAS16], graph neural networks [GZL+23], or reinforcement learning [ACF23].

However, by fitting a user's like, swipe and scroll actions, these algorithms may create radicalization [GHWZ26] and affective polarization [Tör22], overemphasize instinctive preferences while dismissing thoughtful judgments [Kah11], and exacerbate rabbit holes that endanger mental health [AWD+24]. Such considerations are unfortunately too often deprioritized by academic research. For instance, the survey [AHH+25] reads "the main objective of RS is to increase the company's sales." Given the recent judicial rulings in the US and the EU about "addictive social media designs" that include a qualification of some recommendation systems as "dark patterns" [Com26, Reu26, SL26], the algorithms that result from this line of research could be considered to run the risk of producing potentially illegal algorithms.

Conversely, increasing attention has been given to "trustworthy recommender systems" [GLF+25] with greater focus on privacy [LU12], explainability [ZC20], fairness [DU22, WMZ+23], diversity [WZM+24], robustness [ZCS+26] and user control [AFE+21]. Our solution VECHE arguably scores brilliantly in all these dimensions. It naturally provides high explainability, strong fairness between followed users, robustness

to non-followed users, and user control through follow decisions and hyperparameter selections. By upgrading it with Tillé's pivot algorithm **[CITE]**, we can additionally provide diversity. Finally, Section 4.4 discusses how a decentralization of VECHE can provide privacy for following users, and how homomorphic encryption [RAD+78] and differential privacy [Dwo08] could yield privacy for followed users.

To construct VECHE, we built upon inspiring prior work on AI governance. We especially leverage the concept of simulating a vote between the stakeholders' algorithmic representatives introduced by WeBuildAI [LKK+19] for food donation, as well as by [NGA+18] and [FBS+20] for autonomous car dilemmas and kidney donation. Inspired by the more careful psychological analysis of human preferences of [Hoa19, LM23], VECHE also provides a clear distinction between a user's preferences for what the recommendations they receive, and their preferences for what they want their followers to be recommended, Finally, it leverages alternative platforms that welcome alternative recommendation systems, in particular BlueSky [KFG+24] and Tournesol [HBF+25].

3 Veche

Consider a set $[E] \triangleq \{1, 2, \ldots, E\}$ of E entities that may be recommended to Alice. We consider the problem of creating a feed $f \in \mathcal{F}^E$, where

$$\mathcal{F}^E \triangleq \left\{ f \in [E]^S \,\middle|\, S \in \mathbb{N} \wedge |f([S])| = S \right\}. \tag{1}$$

Put differently, a feed f is a ranked subset of $[E]$. Denoting $\textsc{Im}(f) \subset [E]$ the recommended subset, f must define an order of $\textsc{Im}(f)$, i.e. a bijection $|\textsc{Im}(f)| \to \textsc{Im}(f)$. Let $[U]$ be the set of the U users that Alice follows. As stated in the introduction, we propose to construct the feed f by making the U followed users vote for what entities they want to recommend to Alice. More precisely, a stochastic feed based on followed users' votes must be a map

$$\textsc{Ballot}^U \to \Delta(\mathcal{F}^E), \tag{2}$$

where $\textsc{Ballot}$ is a set of allowed ballots and where $\Delta(\mathcal{F}^E)$ is the set of distributions over $\mathcal{F}^E$.

VECHE is a stochastic feed based on followed users' votes, where the set of ballots is $\textsc{Ballots} \triangleq \mathbb{R}^E$. Put differently, in VECHE, each user u may submit any ballot $b_u \in \mathbb{R}^E$, where $b_{ue} \in \mathbb{R}$ corresponds to the extent to which the user u wants entity $e \in [E]$ to be included into Alice's feed. Furthermore, VECHE uses an intermediary ballot aggregation that maps the followed users' ballots $b \in \mathbb{R}^{U \times E}$ to a weight vector $w \in \mathbb{R}^E_{\geq 0}$. We then leverage feed algorithms that guarantee that the probability of recommending $e \in [E]$ is tightly linked to the weight w_e. In principle, VECHE could include any ballot vector aggregation rule $\mathbb{R}^{U \times E} \to \mathbb{R}^E_{\geq 0}$ to transform users' ballots into an entity weight vector. It is important, however, that the mechanism be mathematically well-behaved. Typically, it should bound the influence of any user $u \in [U]$.

3.1 Veche0: Veche in its simplest form

For the sake of exposition, let use first state a simple instantiation of VECHE, which we call VECHE0. We will complexify it in Section 4.2. VECHE0 first considers the simple ballots given by

$$b_{ue} \triangleq \begin{cases} 1, & \text{if } u \text{ posted or reposted } e, \\ -1, & \text{if } u \text{ reported } e, \\ 0, & \text{otherwise.} \end{cases}$$

Then, VECHE0 simply normalize s each user's ballot b_u by the some of its weights, i.e.

$$\tilde{b}_u \triangleq \begin{cases} b_u / \|b_u\|_1, & \text{if } \|b_u\|_1 > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

Next, VECHE0 may simply add up all users' normalized ballots, while demanding nonnegative weights, yielding

$$\mathrm{W}(b) \triangleq \max\left\{0, \sum_{u\in[U]} \tilde{b}_u\right\}. \tag{4}$$

Finally, VECHE0 constructs the feed f by repeated sampling without replacement, where the probability of selecting e is proportional to w_e at each sampling. VECHE0 is detailed in Algorithm 1.

Algorithm 1 VECHE0: A simple instantiation of VECHE

Require: Maximal size S of returned recommendation subsets. Users' ballots $b \in \mathbb{R}^{U\times E}$.

```
b̃_u ← b_u / ‖b_u‖_1                          ▷ Normalization (with convention 0/0 = 0)
w ← max{0, Σ_{u∈[U]} b̃_u}                    ▷ Aggregation of users' votes
E ← {e ∈ [E] | w_e > 0}                      ▷ Filters non-recommendable entities
f ← (⊥, ..., ⊥) ∈ {⊥}^E                      ▷ Instantiate recommendation subset
for i = 1, 2, ..., |E| do                    ▷ Subset sampling without replacement
    p ← w / ‖w‖_1                            ▷ Normalize weights to obtain a probability
    e* ← CHOICE(p)                           ▷ Select entity e with probability p_e
    f_i ← e*                                 ▷ Append selected entity to ranking
    w_{e*} ← 0                               ▷ No replacement: prevent reselecting selected entity
end for
return Feed f of the subset E of recommendable entities.
```

Overall, VECHE0 defines a map $\mathbb{R}^{U\times E} \to \Delta(\mathcal{F}^E)$, which inputs followed users' ballots $b \in \mathbb{R}^{U\times E}$ and returns a probability distribution over rankings of a subset of recommendable entities in $[E]$.

3.2 Main gurantee

Let us formalize the main guarantees of VECHE0, which essentially consist of bounding the maximal impact of a followed user on Alice's feed. As Alice's feed is random, we first need to a metric over probability distribution. In this paper, we will use the classical total variation distance.

Definition 1. *The total variation distance between the probability distributions $\tilde{x}, \tilde{y} \in \Delta(\mathbb{X})$ over $\mathbb{X}$ is the maximal probability distance over measurable subsets $X \subset \mathbb{X}$, i.e.*

$$\mathrm{TVDist}(\tilde{x}, \tilde{y}) \triangleq \sup_{X\subset\mathbb{X} \text{ measurable}} \left|\mathbb{P}\left[x \in X \mid x \sim \tilde{x}\right] - \mathbb{P}\left[y \in X \mid y \sim \tilde{y}\right]\right|.$$

To state our guarantee, we also consider the following indicator, which we show to be a seminorm.

Definition 2. *Define the minimal remaining weights at the k-th recommendations as*

$$\|w\|_{1,k} \triangleq \inf_{\substack{\mathbf{E}\subset[E] \\ |\mathbf{E}|=E-k+1}} \sum_{e\in\mathbf{E}} |w_e|. \tag{5}$$

Note that the minimal weight after 0 recommendations matches the ℓ_1 norm, i.e. $\|w\|_{1,1} = \|w\|_1$.

Proposition 1. *$\|\cdot\|_{1,k}$ is a seminorm.*

Proof. The proof is given in Appendix A. □

Denote $\mathrm{VECHE0}_{1:k}(b) \in \Delta\left(\left([E] \cup \{\perp\}\right)^k\right)$ the distribution of the subset of entities in the top k recommendations. In particular, $e \in \mathrm{VECHE0}_{1:k}(b)$ is the binary event that says whether e was included in the recommendations. For $b \in \mathbb{R}^{U\times E}$ and $u \in [U]$, denote b^{-u} the ballot obtained by removing user $u \in [U]$. Then the following holds.

Theorem 1. *Let $b \in \mathbb{R}^{U \times E}$ and $k \in \mathbb{N}$. If $\|W(b)\|_{1,k} > 0$, then*

$$\text{TVDist}\left(\text{Veche0}_{1:k}(b), \text{Veche0}_{1:k}(b^{-u})\right) \leq k \cdot \|W(b)\|_{1,k}^{-1}. \tag{6}$$

Proof. The proof is given in Appendix B. □

The resilience guarantee may feel underwhelming, as it depends on the cumulative nonnegative weights assigned in the absence of user u. In fact, this highlights the fact that the vulnerability of the recommendation system is increased if most users spend their ballots to oppose the recommendation of some entities, thereby leaving out too few votes in favor of entity recommendation. To clarify the role of negative ballot weights on the security guarantees with respect to a malicious followed user, we derive a corollary of the theorem for nonnegative ballot weights.

Corollary 1. *Assume nonnegative ballots $b \in \mathbb{R}_{\geq 0}^{U \times E}$. Then*

$$\text{TVDist}\left(\text{Veche0}_1(b), \text{Veche0}_1(b^{-u})\right) \leq 1/(U-1). \tag{7}$$

Proof. Assuming nonnegative votes, we have $\|W(b^{-u})\|_1 = \sum_{e \in [E]} \sum_{v \neq u} \tilde{b}_{ve} = \sum_{v \neq u} \|b_v\|_1 = \sum_{v \neq u} 1 = U - 1$. We then apply Theorem 1 with $k = 1$. □

4 Generalizations

In Section 3, we introduce an instantiation of Veche and of algorithmic representatives. However, these simple versions have several shortcomings, which we believe to be important to improve upon for a widespread adoption. Let us now present various generalizations to increase user-friendliness, expressiveness and adaptiveness.

4.1 Ballot biasing

Both Alice and the users she follow could engage in ballot biasing, to better fit Alice's preferences. All below biasing functions are maps $\mathbb{R}^E \to \mathbb{R}^E$, which could either be used by a recommending user $u \in [U]$ (or rather, by their algorithmic representative) to construct their ballot b_u, or by Alice, prior to ballot normalization and aggregation.

4.1.1 Time decay bias

We start with arguably the most important bias, namely favoring more recent activities. Leveraging time biases especially allow us to challenge the chronological feed, though, importantly, we still guarantee fairness between followed users.

Time bias is more reasonably introduced by the recommending user $u \in [U]$. Denoting τ_{ue} the characteristic lifetime of entity e according to user u, and given t_{ue} the date of user u's action on entity e, at time $t \in \mathbb{R}$, we may then define

$$\text{Bias}_e^{time}(b_u | t, t_{ue}, \tau_{ue}, \tau_0, \gamma_\tau) \triangleq b_{ue} \cdot \frac{\tau_{ue} \ln(1 + \tau_{ue}/\tau_0)^{\gamma_\tau}}{(t - t_{ue})^2 + \tau_{ue}^2}. \tag{8}$$

The function has two hyperparameters τ_0, which is a typical lifetime, and γ_τ, which defines whether longer-term entities are favored. The larger γ_τ, the more larger lifetimes will be favored.

Our specific decay bias verifies the following properties. The proofs are postponed to Appendix C.

Proposition 2. *The ballot weight is halved after a time τ has passsed, i.e. $b_{ue}(\tau + t_{ue}) = b_{ue}(t_{ue})/2$.*

	τ_0	γ_τ
Tournesol videos	1 month	1
BlueSky	1 day	-1

Table 1: Suggested decay hyperparameter values (τ_{ue} would be set by default to τ_0).

Proposition 3. *The ratio of the ballot weights of two entities e, f on which the last actions were taken at times t_{ue} and t_{uf} converge to their base ballot weight ratio b^0_{ue}/b^0_{uf} when $t - t_{uf} \gg |t_{ue} - t_{uf}|$ and $t - t_{uf} \gg \tau$. More precisely, we have*

$$\frac{\textsc{Bias}^{time}_e(b_u)}{\textsc{Bias}^{time}_f(b_u)} = \frac{b_{ue}\ln(1+\tau_{ue}/\tau_0)^{\gamma_\tau}}{b_{uf}\ln(1+\tau_{uf}/\tau_0)^{\gamma_\tau}}\left(1 + 2\frac{t_{uf}-t_{ue}}{t-t_{uf}} + O\left(\frac{(t_{uf}-t_{ue})^2}{(t-t_{uf})^2}\right) + O\left(\frac{\tau^2(t_{uf}-t_{ue})}{(t-t_{uf})^3}\right)\right). \tag{9}$$

Proposition 4. *Suppose user u's action times follow a Poisson point process of parameter λ that starts at time 0, and that all entities are given $\tau_{ue} = \tau$. Then, at any time $t \geq 0$ the user's expected sum of ballot weights is bounded as*

$$\forall t \in \mathbb{R}_{\geq 0},\ \mathbb{E}\left\|\textsc{Bias}(b_u|t, t_u, \tau_u, \tau_0, \gamma_\tau)\right\|_1 \leq \frac{\lambda\pi}{2}\cdot\ln(1+\tau/\tau_0)^{\gamma_\tau}. \tag{10}$$

In particular, any new entity is expected to compete with a bounded amount of weights given to other entities, and does not have a vanishing probaility of being pushed by user u.

Proposition 5. *Then the cumulative ballot weight associated to an action over time is given by*

$$\int_{t_{ue}}^{\infty} \textsc{Bias}^{time}_e(b_u|t, t_{ue}, \tau_{ue}, \tau_0, \gamma_\tau)dt = \frac{\pi}{2} b_{ue}\ln(1+\tau/\tau_0)^{\gamma_\tau}. \tag{11}$$

In particular, γ_τ indeed weighs the extent to which long-term or short-term entities should be prioritized.

4.1.2 Other biases

We now discuss other useful biases to inject on ballot weights.

Removing previously recommended entities. While Alice may prefer diverse recommandations, she may also want to previously recommended entities to be recommended again, especially if they are highly reposted by the users she follows. Moreover, we may assume that the more she is recommended an entity, the more time she wants to fly before it is recommended again. We capture this by denoting $n_e(t)$ the number of times an entity is recommended. $t^{last}_e \in \mathbb{R} \cup \{\infty\}$ the time of the entity was recommended and a base delay $\tau_{\circlearrowright}$ between two recommendations. Any user u's ballot $b_u(t)$ at time t is then modified by the non-recommendations of recently recommended entities, as

$$\textsc{Bias}^{\circlearrowright}_e(b_u|t, t^{last}_e, n_e, \tau_{\circlearrowright}, k_{\circlearrowright}) \triangleq b_{ue}\cdot\mathbb{1}\left[t \geq t^{last}_e + \tau_{\circlearrowright}k^{n_e}_{\circlearrowright}\right], \tag{12}$$

where $k_{\circlearrowright}$ is the base of the geometric time delay between two recommendations. This modified ballot is then the one that will be normalized as described above.

Discoverability bias. Alice and/or followed user $u \in [U]$ may want to favor discoverability, by recommending entities that have had fewer views. Let N^{views}_e the number of views of entity $e \in [E]$. We model their discoverability bias preferences by

$$\textsc{Bias}^{\star}_e(b_u|N^{views}_e, N^{\star}_{min}, \gamma_\star) \triangleq b_{ue}\cdot\left(\frac{\log_2(N^{\star}_{min})}{\log_2(N^{\star}_{min}+N^{views}_e)}\right)^{\gamma_\star}, \tag{13}$$

where $N^\star_{min}$ and $\gamma_\star$ are a tunable hyperparameter that encodes the extent to which discoverability is demanded. Entities with $N^\star_e \ll N^\star_{min}$ are treated similarly. Meanswhile, larger values of $\gamma_\star$ will all the more favor entities with few views. With $N^\star_{min} = 100$ and $\gamma_\star = -2$, an entity with 100 views is given 76% of the weight of an entity with 0 views, 3 times more weights than an entity with $10k$ views, and 6.8 times more weights than an entity with 1 million views.

Semantic bias. Alice's aggregation can be customized based on her semantic preferences. There are a myriad of ways to do so, without effect on the principle of unit (or volume-based) influence per followed user. Here we propose an approach based on vector representations, which may be given by a language model. First we need to represent Alice's preferences by a vector embedding $x_{Alice} \in \mathbb{R}^D$. This embedding could be a weighted average of the embeddings of her bio, posts, reposts and liked entities. It could also be the embedding of an explicit search that Alice made. We do not detail how it is obtained. Similarly, we map all entities $e \in [E]$ to a vector $x_e \in \mathbb{R}^D$, and we define the semantic relevancy of entity e for Alice using the cosine similarity. Weighting the ballots with this cosine similarity may then yield

$$\textsc{Bias}^{\measuredangle}_e(b_u|x_e, x_{Alice}, \gamma_\measuredangle) \triangleq b_{ue}\left(1 + \frac{x^T_{Alice}x_e}{\|x_{Alice}\|_2 \cdot \|x_e\|_2}\right)^{\gamma_\measuredangle}, \tag{14}$$

where $\gamma_\measuredangle \geq 0$ determines the degree to which user u wants semantic personalization to weigh in.

Lifetime bias. Alice may prefer trends and news, or she may prefer long-term information. Similarly, a followed user may want to recommend one or another kind of information. We model this as

$$\textsc{Bias}^{\looparrowright}_e(b_u|\tau_{ue}, \tau_{Alice}, \gamma_{\looparrowright}) \triangleq b_{ue} \cdot \left(1 + \left|\ln \frac{\tau_{Alice}}{\tau_{ue}}\right|^2\right)^{-\gamma_{\looparrowright}}. \tag{15}$$

Note that this requires τ_{ue} to be known. If Alice wants to perform lifetime bias, she could query it as part of a Veche protocol extension, or this could be inferred by Alice given a semantic analysis of entity e.

Combining all biases. Both Alice and users $u \in [U]$ may combine the above biases and more. We denote Bias and $\textsc{Bias}_u$ the compositions of the bias functions that they leverage. Table 2 yields suggested bias hyperparameters.

	$\tau_\circlearrowright$	$k_\circlearrowright$	$N^\star_{min}$	$\gamma_\star$	$\gamma_\measuredangle$	$\gamma_{\looparrowright}$
Tournesol videos	1 week	1.2	100	2	1	1
BlueSky	6 hours	2	100	1	2	2

Table 2: Suggested bias hyperparameters for Tournesol and BlueSky

4.2 Ballot aggregation

We now present generalizations of Alice's Veche.

4.2.1 User volumes

Thus far, we assumed that Alice had a binary choice. She could either follow or not follow a given user $u \in [U]$. However, our algorithms generalize naturally to the case where she may tune the volume v_u of user u. The weights would then be given by

$$w \triangleq \max\left\{0, \sum_{w \in [U]} v_u \tilde{b}_u\right\} \tag{16}$$

Time-decaying volume. Denote $t_u \in \mathbb{R}$ the date of Alice's follow of user $u \in [U]$. To account for the tendency between any two users to drift away through time, we propose a decay of the volume v_u as

$$v_u^{\Rsh}(t) \triangleq \frac{v_u \tau_{\Rsh}^2}{\tau_{\Rsh}^2 + (t - t_u)^2}. \tag{17}$$

Evidently, we could add a follow renewal option, to enable Alice to reinitialize a followed user's volume.

Updating volumes based on likes. While we could allow Alice to tune by hand the volume of the users she follows, she might prefer to use implicit signals to do this indirectly and effortlessly. The *like* and *dislike* signals could arguably be used to do so. However, to prevent a "like-lock-in", where an old like is still affecting Alice's recommendations years later, without her being aware of it, we could consider a time discount on each signal. Let us denote $l_n \in \mathbb{R}$ the n-th like action (e.g., $+1$ for like, -1 for dislike), t_n the time of the action, e_n the entity the action was made on. We may then consider that the like action on a user volume refers to the extent to which the user u wanted to add e_n to Alice's recommendation feed, which is given by $\tilde{b}_{nu} \triangleq \tilde{b}_{ue_n}(t_n)$. We may derive the impact of the first N actions on user volumes by accounting for the like intensity l_n and for a time discount, and by adding all such influences. Demanding non-negative volumes then yields

$$v_u(t) \triangleq \max\left\{0, v_u^{\Rsh}(t) + \alpha_{\heartsuit} \sum_{n=1}^{N} l_n \frac{\tilde{b}_{nu}}{\left\|\tilde{b}_n\right\|_1} \frac{\tau_{\heartsuit}^2}{\tau_{\heartsuit}^2 + (t - t_u)^2}\right\}, \tag{18}$$

where $\tau_{\heartsuit}$ is the characteristic lifetime of a like action, and where $\alpha_{\heartsuit}$ could be tuned by Alice to select the influence of her likes on volume modifications. In practice, relevant user interfaces and notifications could be used to inform Alice of the impact of her like actions on the volumes, and to allow her to adjust the volumes independently from the like actions.

We stress that, if our algorithms are used as stated here, Alice's like actions would only affect Alice's recommendations, and not others' recommendations. This is in sharp contrast with Alice's repost actions, which has no effect on Alice's recommendations, but will affect the recommendations to the accounts that follow Alice. Put differently, our proposed implementation formalized the following principle.

Likes inform on what a consumer wants for themselves.
Reposts and reports inform on what they want for their followers.

While we recognize that this may not be how social media consumers currently interprets likes and reposts, we strongly believe that a clarification of the meanings of their signals is useful to help them contribute to a better governed information space.

Adding non-followed users. Alice could also want to interact with users that she does not follow, and in particular to allow them to directly affect her feed. Indeed, she may thereby engage with responses they provide to her message, or she may be opened to receive recommendations of entities in which she is mentioned. In particular, she could want to make such responses and mentions part of a main feed, to avoid having another response/mention feed with notifications that she could regard as addictive, if not part of an undesirable dark pattern.

Nevertheless, we recognize that Alice may not want such interactions, especially if they involve harassment. In fact, she should arguably be given the possibility to set the maximal "volume" v_{max} of a non-followed user, and to limit the total amount of interactions with non-followed accounts. Denote $\hat{\mathbf{U}}$ the set of such interacting non-followed accounts. We consider that their cumulative voting right must be at most proportional to U. Namely, denoting $\hat{v}_{all} \geq 0$ the proportionality coefficient, their cumulative voting right must not exceed $\hat{v}_{max} U$. We then spread this total volume to the accounts $u \in \hat{\mathbf{U}}$, with the constraint

that none should have more than the volume v_{max} that followed users are given by default, i.e. we set

$$\forall u \in \hat{\mathbf{U}},\ v_u \triangleq \min\left\{\hat{v}_{max}, \hat{v}_{all}\frac{U}{\left|\hat{\mathbf{U}}\right|}\right\}. \tag{19}$$

Hyerparameters. Table 3 lists suggested volume hyperparameters.

	$\tau_{\Rsh}$	$\tau_{\heartsuit}$	$\alpha_{\heartsuit}$	$\hat{v}_{max}$	$\hat{v}_{all}$
Tournesol videos	5 years	1 year	0.1	1/2	1/5
BlueSky	1 year	1 month	0.1	1/2	1/5

Table 3: Suggested volume hyperparameters for Tournesol and BlueSky

4.2.2 Ballot normalization

In the basic instantiation of VECHE, all signals (posts, reposts, reports...) are treated similarly, and are normalized by spreading the total ballot weights. We highlight the limits of this approach, and show extensions.

Limit the voice of decidedly silent users. A user $u \in [U]$ may be so inactive and they may prefer not recommending anything, rather than recommending entities that Alice may not like, and which could deteriorate her experience. In fact, Alice could even be bothered by u, and decide to reduce u's volume. To mitigate this, we simply allow $\tilde{b}_u$ to be of smaller norm, i.e. we set

$$\tilde{b}_u \triangleq b_u \min\left\{1, \|b_u\|_1^{-1}\right\}. \tag{20}$$

Thereby, by reporting b_u with $\|b_u\|_1 < 1$, user u guarantees $\left\|\tilde{b}_u\right\|_1 < 1$.

Ballot ℓ_q normalization. Thus far, we used the normalization $\tilde{b}_u \triangleq b_u / \|b_u\|_1$. This ℓ_1-norm has the advantage of viewing $\tilde{b}_u$ as a way for user u to allocate their volume v_u across the entities $e \in [E]$. However, this approach has been criticized for its efficiency and robustness guarantees [LW18], who instead call for *quadratic voting*. This would boil down to replacing the ℓ_1-norm with an ℓ_2-norm. More generally, for any $q \in [1, \infty]$, we could define the ℓ_q-normalization

$$\tilde{b}_u \triangleq b_u \|b_u\|_q^{-1}. \tag{21}$$

Larger values of q also tend to favor users that spread their ballot weights b_{ue} across more entities. Assuming that ballots are derived directly from actions as has been proposed earlier, this would incentivize users' participations on the social media. While it may be desirable to provide such incentives to some extent, this could also imply negative impacts, such as over-valuing hyperactive users or mental health harm to creators. Arguably, $q = 2$ could strike a good balance.

Different normalizations for different signals. We believe it to also be interesting, for various reasons, to normalized different signals differently. The main reason for this is that a user u might want above all to have their posts recommended, which may trigger concerns of mass self-promotion, and a lack of effort to evaluate others' posts. To avoid the possibility that other signals (reposts, reports) will harm a user u's posts, we may simply normalize each signal differently. Namely, denote $\mathbf{E}_u^{post}, \mathbf{E}_u^{\uparrow}, \mathbf{E}_u^{\downarrow}$ the subsets of entities,

which are respectively posted, evaluated positively or evaluated negatively by user u (assuming that entities posted by u cannot be evaluated). Define

$$b_{ue}^{post} \triangleq b_{ue} \mathbb{1}\left[e \in \mathbf{E}_u^{post}\right], \; b_{ue}^{\uparrow} \triangleq b_{ue} \mathbb{1}\left[e \in \mathbf{E}_u^{\uparrow}\right], \; b_{ue}^{\downarrow} \triangleq b_{ue} \mathbb{1}\left[e \in \mathbf{E}_u^{\downarrow}\right]. \tag{22}$$

Assuming that Alice wants to weight positive and negative evaluations with weights $\alpha_\uparrow$ and $\alpha_\downarrow$ respectively (compared to posts with unit weight), she could normalize the ballots as

$$\tilde{b}_u \triangleq b_u^{post} \left\|b_u^{post}\right\|_1^{-1} + \alpha_\uparrow b_u^{\uparrow} \left\|b_u^{\uparrow}\right\|_1^{-1} + \alpha_\downarrow b_u^{\downarrow} \left\|b_u^{\downarrow}\right\|_1^{-1} \tag{23}$$

with the convention $0 \cdot 0^{-1} = 0$. Larger values of both $\alpha_\uparrow$ and $q_\uparrow$ would incentivize users' evaluations and promotions of other users' entities, which could also be viewed as a way to reward content evaluation. Meanwhile, larger values of $\alpha_\downarrow$ and $q_\downarrow$ would make it easier for users to prevent harmful entities from being recommended. Typically, with $\alpha_\downarrow = 6$ and $\alpha_\uparrow = 2$, an entity would have to be posted or reposted by at least two times as many users than the number of users that have reported it. This may be desirable to prevent the spread of divisive entities, and to instead favor more consensual entities.

Combining all ballot normalizations. Considering regularization norms q_{post}, $q_\uparrow$ and $q_\downarrow$ and aggregating all normalizations listed above yield

$$\tilde{b}_u \triangleq b_u^{post} \min\left\{1, \left\|b_u^{post}\right\|_{q_{post}}^{-1}\right\} + \alpha_\uparrow \min\left\{1, b_u^{\uparrow} \left\|b_u^{\uparrow}\right\|_{q_\uparrow}^{-1}\right\} + \alpha_\downarrow \min\left\{1, b_u^{\downarrow} \left\|b_u^{\downarrow}\right\|_{q_\downarrow}^{-1}\right\}. \tag{24}$$

Table 4 provides suggested hyperparameters.

	q_{post}	$\alpha_\uparrow$	$q_\uparrow$	$\alpha_\downarrow$	$q_\downarrow$
Tournesol videos	1	1	2	4	2
BlueSky	1	1	2	4	2

Table 4: Suggested ballot aggregation hyperparameters for Tournesol and BlueSky

4.2.3 Diversifying the recommended feed

Algorithm 2 Tillé's pivot algorithm

Require: Weights $w \in \mathbb{R}_{\geq 0}^E$, $\textsc{Scheduler} : 2^E \to \Delta(E^2)$.
  $\mathbf{E} \leftarrow \{e \in [E] \,|\, w_e > 0\}$ ▷ Initialize candidates as set of recommendable entities
  $f \leftarrow \text{NewList}()$ ▷ Initialize feed
  while $|\mathbf{E}| > 0$ **do**
    $e_1, e_2 \leftarrow \textsc{Scheduler}(\mathbf{E})$ ▷ Draw entities, one of which will be eliminated
    if $\textsc{Bernoulli}(w_{e_1}/(w_{e_1} + w_{e_2})) = 0$ **then** ▷ If e_1 loses against e_2
      $e_1, e_2 \leftarrow e_2, e_1$ ▷ Switch order
    end if
    $w_{e_1} \leftarrow w_{e_1} + w_{e_2}$ ▷ Winning entity gains loser's weight
    $\mathbf{E} \leftarrow \mathbf{E} - \{e_2\}$ ▷ Remove loser
    Prepend e_2 to feed f ▷ Prepends to feed
  end while
  return feed f.

Feed diversification is often considered preferrable, to prevent rabbit holes. We show how Veche can achieve this, by leveraging Tillé's pivot algorithm [CITE]. The algorithm is described in Algorithm 2. Crucially, the main guarantee still holds. We state it below, and prove it in Appendix D.

Theorem 2. *Assume* $\|\mathrm{W}(b)\|_1 > 1$. *Then*

$$\mathrm{TVDist}\left(\mathrm{TilleVeche}_1(b), \mathrm{TilleVeche}_1(b^{-u})\right) \leq \left(\left\|\mathrm{W}(b^{-u})\right\|_1 - 1\right)^{-1}. \tag{25}$$

Tillé's pivot algorithm depends on a *pivot scheduler* Scheduler, which can increase the diversity within the recommended subset. We propose to instantiate a scheduler by considering a similarity measure between any pairs of entities e and f given by the Pearson correlation ρ_{ef} between the ballot weights $(b_{ue})_{u\in[U]}$ and $(b_{uf})_{u\in[U]}$. Denoting $\bar{b}_e \triangleq (\sum_u b_{ue})/U$, this yields

$$\rho_{ef} \triangleq \frac{\sum_u (b_{ue} - \bar{b}_e)(b_{uf} - \bar{b}_f)}{\sqrt{\sum_u (b_{ue} - \bar{b}_e)^2}\sqrt{\sum_u (b_{uf} - \bar{b}_f)^2}}. \tag{26}$$

The correlation is maximal when two entities are posted by the same user, with no interaction from any other users. The following proposition holds about correlations. The proof is given in Appendix D.

Algorithm 3 Scheduler based on ballot correlations

Require: Ballots $b \in \mathbb{R}^{U\times E}$, subset $\mathbf{E} \subset [E]$.
  $\mathrm{Pairs} \leftarrow \left\{(e,f) \in \mathbf{E}^2 \,\middle|\, e < f\right\}$
  for $(e,f) \in \mathrm{Pairs}$ **do**
    $\rho_{ef} \leftarrow \mathrm{PearsonCorr}\left(b_{\cdot e}, b_{\cdot f}\right)$ ▷ Estimate similarity
    $\lambda_{ef} \leftarrow \left(\frac{1+\rho_{ef}}{1-\rho_{ef}}\right)^{\gamma_\lambda}$ ▷ Derive pair sampling weight
  end for
  $P_\infty \leftarrow \{(e,f) \mid \lambda_{ef} = \infty\}$ ▷ Pairs that must be deterministically prioritized
  if $|P_\infty| \geq 1$ **then**
    $(e,f) \leftarrow \mathrm{Uniform}(P_\infty)$ ▷ Uniform sample over prioriy pairs
  else
    $(e,f) \leftarrow \mathrm{Choice}\left(\mathrm{Pairs}, \lambda / \sum_{(e,f)\in\mathrm{Pairs}} \lambda_{ef}\right)$ ▷ Sample proportionally to λ_{ef}
  end if
  return (e,f)

Given a subset $\mathbf{E}$ and two distinct entities $e, f \in \mathbf{E}$, we assign a weight p_{ef} to (e,f) given by

$$\lambda_{ef} \triangleq \left(\frac{1+\rho_{ef}}{1-\rho_{ef}}\right)^{\gamma_\lambda} \in \mathbb{R}_{\geq 0}, \tag{27}$$

Note that the map $\rho \mapsto \frac{1+\rho_{ef}}{1-\rho_{ef}}$ is a bijection $[-1,1] \to \mathbb{R}_{\geq 0} \cup \{\infty\}$. We now construct Scheduleras a sampler of pairs of distinct entities, such that the probability of drawing (e,f) be proportional to λ_{ef}. Evidently, a little care is needed when some values λ_{ef} are infinite. Essentially, we solve this issue by drawing uniformly randomly among pairs with infinite λ_{ef} if some exist, and by drawing proportionally to λ_{ef} otherwise. Algorithm 3 details the algorithm, where $\mathrm{Choice}(A,p)$ randomly selects an object a of A with probability p_a. Table 5 provides suggested values for the diversity hyperparameter γ_λ. Larger values imply greater diversity.

As stated, Scheduler is unfortunately quadratic in E. To accelerate the algorithm, a heuristic could consist of uniformly randomly selecting an entity $e \in \mathbf{E}$, and then selecting a similar entity within a random subset of $\mathbf{E}$ of fixed small size S. This would reduce the complexity to $O(ES)$.

4.2.4 More radical generalizations

While our generalizations already produce a certain number of hyperparameters, thereby already adding a significant amount of complexity that hampers transparency for layman users, they could be considered

	$\gamma_\curlywedge$
Tournesol	1
BlueSky	2

Table 5: Suggested diversity hyperparameters for Tournesol and BlueSky

still to restricted to a computational social choice scientist. In particular, high-dimensional social choice theory and AI security have introduced a large toolbox of vector aggregation mechanisms, which could be worth exploring to evaluate whether the additional complexity could be justified through more desirable mathematical guarantees. In Appendix E, we discuss a way to use the geometric median and its advantages, especially as a way to account for algorithmic representatives' uncertainties on what the users they represent would prefer. However, a more thorough investigation of better-behaved ballot aggregations is left for future work.

4.3 Algorithmic representative generalizations

We now discuss generalizations of followed users' algorithmic representatives.

Liquid democracy. A user $u \in [U]$ could want to relay another user v's posts, even when u has not explicitly reposted v's posts. This allows u to systematically promote v, without having to check whether v has a new post that u could repost. This approach can be naturally interpreted by liquid democracy. By setting a relay volume $r_{uv}(t) \in \mathbb{R}$ at time t, user u's ballot powered by liquid democracy could then be derived as

$$b^{\rightsquigarrow}_{ue}(t) \triangleq b_{ue}(t) + \sum_{v \in \mathbf{U}^{\rightsquigarrow}_u} r_{uv}(t) b_{ve}(t), \tag{28}$$

where $\mathbf{U}^{\rightsquigarrow}_u$ is the set of users v that user u relays. Note that negative values could also be used to vote against the actions of v. While liquid democracy could thereby provide a new other way for users to support creators, we caution about the risk that this could be marketized, as some creators may then pay some users to systematically promote their ballots. Additionally, a user u might forget about past relays they set up, with which they may no longer agree, especially if v's positions have evolved. To account for this risk, we would urge for clear monitoring interfaces of a user's relays, as well as a decay of $r_{uv}(t) \to 0$ as $t \to \infty$ in the absence of new signals that u wants to relay v.

Including additional signals such as ratings and comparisons. A user $u \in [U]$ may also want to better monitor the ballots b_u sent by their algorithmic representatives, as presented in Figure 1, and to modify them based on their own preferences. This could be done through a better control of the initial values b^0_u, or through comparison interfaces that allows the user to report that they prefer recommending e over f, i.e. they want $b_{ue}(t) \geq b_{uf}(t)$. Finally, a user's preferences could be generalized to entities they have not evaluated or reposted. We refer to [FFHV24, FGH$^+$26] for ressources on how this could be achieved. While such approaches may be valuable to allow users u to better control their algorithmic representatives, because of the added complexity they imply, we keep further considerations out of the scope of this paper.

4.4 Future developments of Veche to improve resilience

Extended ballots and retrocompatibility. By construction, Veche accepts any ballots $b_u \in \mathbb{R}^E$. In practice, for communication efficiency, it may simply collect partial ballots $b_u \in \mathbb{R}^{<E}$, where $\mathbb{R}^{<E} \triangleq \bigcup_{\mathbf{E} \subset [E]} \mathbb{R}^{\mathbf{E}}$. The missing coordinates may simply be filled as equal to zero. In future versions, Veche could also accept lifetimes $\tau_u \in \mathbb{R}^{<E}_{>0}$, with retrocompatibility defining a default value for unspecified lifetimes

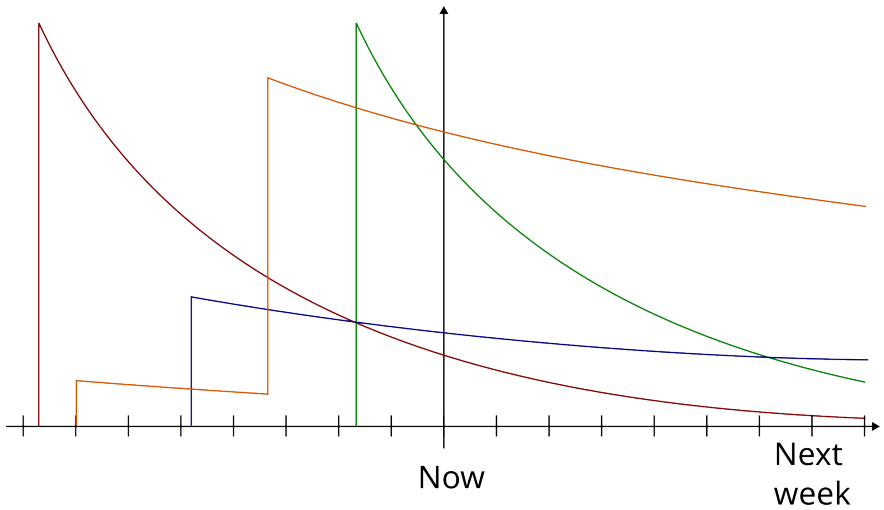

Figure 1: Interface for a Veche recommender to visualize their recommendations. Each line corresponds to the recommendation of an entity. The dark red and dark green lines correspond to a lifetime of 4 days. The dark blue corresponds to a lifetime of a month. The orange line is a relay, which was then heavily positively evaluated for massive recommendations, with a lifetime of a month. At the present time, the orange entity is the most recommended, ahead of the green, blue and red. However, due to diverging lifetimes, the blue entity will eventually be more recommended.

that may typically match the hyperparameter τ_0. Similarly, as is done in Solidago [HBC+22], Veche could additionally accept the left and right uncertainties $\Delta_u^{left}, \Delta_u^{right} \in \mathbb{R}_{\geq 0}^{<E}$, with $\Delta_{ue}^{left} = \Delta_{ue}^{right} = 0$ as the default values for specified ballots b_{ue}, and $\Delta_{ue}^{left} = \Delta_{ue}^{right} = \infty$ if b_{ue} was not specified. Finally, as discussed in E, Veche could consider a hyperparameter $\nu \in \mathbb{R}_{\geq 0}^{<E}$, where ν_e reports the extent to which user u cares about affecting the weight w_e. Its default value could be $\nu_e = 1$.

Interoperability. By construction, Veche is highly interoperable. Followed user $u \in [U]$ may use any algorithmic representative, as long as it returns a ballot $b_u \in \mathbb{R}^E$ at any given time $t \in \mathbb{R}$. Alice could also leverage generalizations or alternatives to Veche, as long as they leverage similar ballots. As we will see, practical computational constraints may however restrict the designs of the algorithmic representatives and ballot aggregations.

Decentralization. Alice could host her own Veche server. At least in principle, Veche servers have no need to be hosted on the same machine. Additionally, in principle, all followed users $u \in [U]$ could have their own algorithmic representative servers. Upon Alice's request for a new feed, Alice's Veche server would query ballots from the followed users' algorithmic servers. Once ballots are collected, the Veche server would construct a feed, which would be sent to Alice.

While this approach highlights the interoperability of Veche, and the sovereignty this may allow, it may be too slow in practice, especially if Alice expects an answer within a fraction of a second. In particular, a dilemma would arise for Alice's Veche server, between systematically waiting for all ballots to be received, and halting before all are received, at the risk of systmatically discriminating slow algorithmic representative servers.

To provide better performance guarantees, followed users may be asked to upload their algorithmic representatives to Alice's Veche server. This is where the name Veche takes all its meaning. Alice's Veche server would thereby be an Assembly of algorithmic representatives. Upon Alice's feed query, the Veche server would execute the algorithmic respresentatives' codes, thereby yielding ballots b, which the Veche server can turn into a feed.

Computational time constraints still hold however in this setting, as the VECHE is still required to answer within a fraction of a second. This implies that each algorithmic representative must have a fast running time, upon query at any point.

Verifiability. In practice, Alice will more likely default to a VECHE server she does not administer, rather than setting up her own VECHE server. Similarly, followed users may leverage algorithmic representatives managed by third parties. To provide Alice with guarantees that her recommendations were indeed obtained by running VECHE with her followers' algorithmic representatives, the VECHE server could leverage verifiable computing, and in particular SNARKs, to prove the correctness of its computations. Similarly, the VECHE server could provide proofs to the followed users that their algorithmic respresentatives' ballots were correctly accounted for, and that their algorithmic representatives ran the correct computations to recommend what the followed users intended to recommend.

Privacy. A followed user could prefer to hide their recommendations from Alice's VECHE server, and perhaps even from Alice. Typically, the user may want to spread a news, without Alice knowing the identity of the person who wanted the news to be shown to her. Similarly, Alice could want to hide her feed, or the list of her followed users, from her VECHE server. Providing such privacy guarantees is however a nontrivial open problem, which we leeave to future work.

Governance. We stress that the hyperparameters of the aggregation are under Alice's control. At least in a setting of algorithmic pluralism where each consumer (like Alice) is given greater control over the recommendations they receive, she will be able to tune herself the impact of the different sources of influence. Having said this, we also stress that, in practice, most consumers do not engage in algorithmic reconfiguration, so that the default value chosen for a widely used recommendation system will heavily influence the dynamics of consumption, production and evaluation of the entities that are communicated through the recommendation system. We believe that an appropriate governance for the selection of the default recommendation system is of the utmost importance.

Additionally, as we discuss it in Section 5, VECHE is likely to generate heavily biased filter bubbles, as any two consumers' lists of followed users may greatly differ. To prevent induced polarization risks, we believe that additional coordinated measures should be considered, to promote uniting entities of public interest [HBF+25].

Finally, as proposed in [HBC+22], the recommendation ecosystem should arguably avoid the emergence of *mass influencers*, i.e. users who can massively bias the diffusion of entities, to prevent the massive spread of a dangerously misleading entity. This could be done by limiting the cumulative volume of any influencer. Better yet, as discussed in [HBC+22], a *presumption of non-recommendability* could be enforced, i.e. an entity would require approval from sufficiently many entities to obtain the right to be recommended more than 1,000 times per day. Proposing mechanisms to provide such guarantees is however beyond the scope of the paper, but represents an exciting future research direction.

5 Evaluating Veche on four pillars

The Tournesol association proposed to evaluate digital democracy solutions based on 4 digital democracy pillars. According to these pillars, solutions must be:

Transparent: The solutions must be free and open-source, but also explainable and verifiable.

Secure: The solutions must be resilient to coordinated attacks, especially through fake accounts. When relevant, they should also allow for private weighing in (like voting in democracy).

Thoughtful: The solutions must value informed and thoughtful judgments, as opposed to rushed, emotional and instinctive actions.

	Transparent	Secure	Thoughtful	Uniting	UX
Proprietary engagement maximization	E	E	E	E	D
Matrix factorization	C	E	D	E	C
Reverse chronological	A	D	D	E	E
VECHE0	A	C	D	E	D
Tournesol-VECHE	B	C	D	D	C
BlueSky-VECHE	B	C	D	D	C
VECHE + Common*	B	B	B	B	C

Table 6: Evaluating Veche on Tournesol's four digital democracy pillars

Uniting: The solutions must unite people, by being inclusive and by contributing to a shared reality.

We refer to `tournesol.app/manifesto` for more justifications of the pillars. We added a fifth evaluation based on user experience. In this section, we evaluate the leading recommendation algorithms along the four pillars, as well as VECHE0, Tournesol-VECHE and BlueSky-VECHE. The latter two include the generalizations discussed in Section 4, with the hyperparameters provided in the various tables of the Section. In particular, both leverage diversification (which allows to be more uniting, as it values diverging viewpoints within the community of followed users). Table 6 summarizes our analysis. Its last line corresponds to the inclusion of common-ground recommendations, as discussed in the following paragraph.

Reinforcing unity. Given this analysis, we actually do not believe that VECHE, nor any follow-based algorithms, should be viewed as a societally satisfactory solution. In particular, such algorithms are inherently facilitating filter bubbles, which may divide societies and harm democracies. This will be even more the case if algorithmic representatives customize their ballots based on the users they target. To construct a common information ground that better unites citizens, we believe that a fraction of the recommendations, say 30%, should instead result from a social choice mechanism that involves all citizens (not only followed accounts), and that yields non-personalized recommendations. Our algorithms may in fact be instrumental to both provide to the remaining 70% of personalized recommendations and to inspire mechanisms that respond to the need for common ground.

6 Experiments

Run code.

7 Conclusion

We believe that our algorithms must be viewed as the preferred default follow-based recommendation algorithm, as opposed to the reverse chronological feed that provides a poor user experience, favors hyperactive users and incentivizes fast and massive content creation. In particular, its practical transparency is a lot more limited than what might appear intuitively. Unless a user follows very few accounts and can read through everything they post and repost, the reverse chronological feed actually removes from their information exposure a lot of information, and the user does not have a lot of mechanism to understand and control the influences they undergo.

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A Proof that $\|\cdot\|_{1,k}$ is a semi-norm (Proposition 1)

Proof. Absolute homonogeneity $\|\lambda w\|_{1,k} = |\lambda| \cdot \|w\|_{1,k}$ is straightforward. We also have

$$\begin{aligned}\|w + w'\|_{1,k} &= \inf_{\substack{\mathbf{E}\subset[E]\\|\mathbf{E}|=E-k+1}} \sum_{e\in\mathbf{E}} |w_e + w'_e| \leq \inf_{\substack{\mathbf{E}\subset[E]\\|\mathbf{E}|=E-k+1}} \left\{\sum_{e\in\mathbf{E}} |w_e| + \sum_{e\in\mathbf{E}} |w'_e|\right\} \\ &\leq \inf_{\substack{\mathbf{E}\subset[E]\\|\mathbf{E}|=E-k+1}} \left\{\sum_{e\in\mathbf{E}} |w_e|\right\} + \inf_{\substack{\mathbf{E}'\subset[E]\\|\mathbf{E}'|=E-k+1}} \left\{+\sum_{e\in\mathbf{E}'} |w'_e|\right\} = \|w\|_{1,k} + \|w'\|_{1,k}\,.\end{aligned}$$

Note however that, while clearly nonnegative, $\|\cdot\|_{1,k}$ is not positive definite. Indeed, for $k \geq 2$, vectors $\mathbf{e}$ of the canonical basis verify $\|\mathbf{e}\|_{1,k} = 0$. □

B Proof of Veche0's resilience (Theorem 1)

This section provides a proof of a generalization of Theorem 1, which bounds the maximal impact of a followed user. More precisely, we consider a slightly generalized setting with user volumes. Define the weights as

$$\mathrm{W}(v, b) \triangleq \max\left\{0, \sum_{u\in[U]} v_u b_u / \|b_u\|_1\right\}. \tag{29}$$

Theorem 3. *Let $v \in \mathbb{R}^U_{\geq 0}$, $b \in \mathbb{R}^{U\times E}$ and $k \in \mathbb{N}$. If $\|W(v, b)\|_{1,k} > 0$, then*

$$\textsc{TVDist}\left(\textsc{Veche0}_{1:k}(v, b), \textsc{Veche0}_{1:k}(v', b),\right) \leq \|v - v'\|_1 \sum_{i=1}^{k} \|\mathrm{W}(v, b)\|^{-1}_{1,i}\,. \tag{30}$$

Note that when $\|v - v'\|_1 > \left(\sum \|\mathrm{W}(v, b)\|^{-1}_{1,k}\right)^{-1}$, the bound becomes trivial as, by definition, the total variational distance lies between 0 and 1. This will hold in particular if $\|v - v'\|_1 > \sum \|\mathrm{W}(v, b)\|_{1,k}$, i.e. the shift in voting right allocations is larger than the worst-case weight allocation after $k - 1$ recommendations. It is also noteworthy that the obvious inequality $\|w\|_{1,i} \leq \|w\|_{1,j}$ for $i < j$ implies that the right-hand side is itself upper-bounded by $k\,\|v - v'\|_1\,\|\mathrm{W}(v, b)\|^{-1}_{1,k}$.

Proof of Theorem 1. Theorem 1 is easily derived from Theorem 3 as the special case where $v \triangleq (1)_{u\in[U]} \in \mathbb{R}^U_{\geq 0}$, and $v'_x \triangleq v_x \mathbb{1}[x \neq u]$ for all $x \in [U]$, i.e. v' removes user u's voting right. □

Relation with Lipschitz resilience. Theorem 3 looks similar to Lipschitz resilience [AGHV24]. Lipschitz resilience demands that the social mechanism $\mathbb{R}^U_{\geq 0} \times \mathbb{X}^U \to \mathbb{Y}$ be Lipschitz continuous as a function of users' voting rights (using the ℓ_1 norm as a metric for voting rights). In our context, we would have $\mathbb{X} \triangleq \mathbb{R}^E$ (user ballots) and $\mathbb{Y} \triangleq \Delta(\mathcal{F}^E)$ with the TVDist metric. Theorem 3 does not guarantee Lipschitz resilience, because our right-hand side is a function of $\|\mathrm{W}(v, b)\|_{1,i}$ for $i \in [1, k]$, which depends on both v and b. This highlights two aspects of Veche0.

First, as already discussed in the main part of our paper, the resilience of Veche0 is weakened by negative ballot weights, as preventing the recommendations of many entities facilitate the arecommendations of other entities. This could be mitigated by the prior identification of definitely safe entities to recommend.

Second, the dependence on v simply highlights the fact that if Alice only follows a single user, then her recommendations will be heavily influenced by this single user. This dependency risk can be mitigated by encouraging Alice to multiply and diversify her list of followed accounts. It could also be mitigated once again by a fallback to safe entities to recommend, when her follow list is insufficiently long.

B.1 Useful lemmas to prove Theorem 3

We first prove a lemma.

Lemma 1. *For all $v \in \mathbb{R}^U_{\geq 0}$ and $b \in \mathbb{R}^{U \times E}$, $\|\mathrm{W}(v,b) - \mathrm{W}(v',b)\|_1 \leq \|v - v'\|_1$.*

Proof. Now note that, for all entity $e \in [E]$, we can bound

$$|w_e - w'_e| = \left| \max\left\{0, \sum_{u \in [U]} v_u \tilde{b}_{ue}\right\} - \max\left\{0, \sum_{u \in [U]} v'_u \tilde{b}_{ue}\right\} \right|. \tag{31}$$

Using the fact that $z \mapsto \max\{0, z\}$ is clearly 1-Lipschitz continuous, we derive

$$|w_e - w'_e| \leq \left| \sum_{u \in [U]} v_u \tilde{b}_{ue} - \sum_{u \in [U]} v'_u \tilde{b}_{ue} \right| = \left| \sum_{u \in [U]} (v_u - v'_u) \tilde{b}_{ue} \right| \leq \sum_{u \in [U]} |v_u - v'_u| \cdot \left| \tilde{b}_{ue} \right|. \tag{32}$$

Adding up over $e \in [E]$ then yields

$$\|w - w'\|_1 = \sum_{e \in [E]} |w_e - w'_e| \leq \sum_{e \in [E]} \sum_{u \in [U]} |v_u - v'_u| \cdot \left| \tilde{b}_{ue} \right| \tag{33}$$

$$= \sum_{u \in [U]} |v_u - v'_u| \sum_{e \in [E]} \left| \tilde{b}_{ue} \right| = \sum_{u \in [U]} |v_u - v'_u| \underbrace{\left\| \tilde{b}_u \right\|_1}_{=1} = \|v - v'\|_1, \tag{34}$$

which yields the lemma. □

Lemma 2. *For $w \in \mathbb{R}^E_{\geq 0}$ with $\|w\|_1 > 0$, define $\tilde{e}(w) \in \Delta([E])$ the distribution of an element $e \in [E]$ selected, if it is drawn with probability $w_e / \|w\|_1$. Then, for any $w, w' \in \mathbb{R}^E_{\geq 0}$ with $\|w\|_1 > 0$ and $\|w'\|_1 > 0$,*

$$\mathrm{TVDist}(\tilde{e}(w), \tilde{e}(w')) \leq \frac{\|w - w'\|_1}{\|w\|_1}. \tag{35}$$

Proof. Given that $\|w\|_1 > 0$ and $\|w'\|_1 > 0$, the distributions of $\tilde{e}(w)$ and $\tilde{e}(w')$ are discrete probabilities in $\Delta([E])$. It is well-known that the total variation distance for discrete probabilities is given by

$$\mathrm{TVDist}(\tilde{e}(w), \tilde{e}(w')) = \frac{1}{2} \sum_{e^* \in [E]} \left| \mathbb{P}\left[e = e^* \mid e \sim \tilde{e}(w)\right] - \mathbb{P}\left[e = e^* \mid e \sim \tilde{e}(w')\right] \right| \tag{36}$$

$$= \frac{1}{2} \sum_{e \in [E]} \left| \frac{w_e}{\|w\|_1} - \frac{w'_e}{\|w'\|_1} \right|. \tag{37}$$

We now leverage the triangle inequality, yielding

$$\mathrm{TVDist}(\tilde{e}(w), \tilde{e}(w')) = \frac{1}{2} \sum_{e \in [E]} \left| \frac{w_e}{\|w\|_1} - \frac{w'_e}{\|w\|_1} + \frac{w'_e}{\|w\|_1} - \frac{w'_e}{\|w'\|_1} \right| \tag{38}$$

$$\leq \frac{1}{2} \sum_{e \in [E]} \frac{|w_e - w'_e|}{\|w\|_1} + \frac{1}{2} \sum_{e \in [E]} \left| \left(\frac{1}{\|w\|_1} - \frac{1}{\|w\|_1} \right) w'_e \right| \tag{39}$$

$$\leq \frac{1}{2\|w\|_1} \sum_{e \in [E]} |w_e - w'_e| + \frac{|\|w'\|_1 - \|w\|_1|}{2\|w\|_1 \cdot \|w'\|_1} \sum_{e \in [E]} |w'_e| \tag{40}$$

$$= \frac{\|w - w'\|_1 + |\|w\|_1 - \|w'\|_1|}{2\|w\|_1}. \tag{41}$$

Again, by the triangle inequality, we have $\|w - w'\|_1 \geq |\|w\|_1 - \|w'\|_1|$. This then yields

$$\text{TVDist}(\tilde{e}(w), \tilde{e}(w')) \leq \frac{|\|w\|_1 - \|w'\|_1| + \|w - w'\|_1}{2\|w\|_1} \leq \frac{\|w - w'\|_1}{\|w\|_1}, \tag{42}$$

which concludes the proof. □

B.2 Total variation distance of joint distribution

For any distribution $\tilde{x}$ on a finite set $\mathbb{X}$, we define $\text{Supp}(\tilde{x}) \triangleq \{x \in \mathbb{X} \mid \mathbb{P}[\tilde{x} = x] > 0\}$ its support.

Lemma 3. *Consider two finite sets $\mathbb{X}_1$ and $\mathbb{X}_2$. Let $\tilde{x}, \tilde{x}' \in \Delta(\mathbb{X}_1 \times \mathbb{X}_2)$. Then*

$$\text{TVDist}(\tilde{x}, \tilde{x}') \leq \text{TVDist}(\tilde{x}_1, \tilde{x}_1') + \sup_{x_1 \in \text{Supp}(\tilde{x}_1) \cup \text{Supp}(\tilde{x}_1')} \text{TVDist}(\tilde{x}_2 | x_1, \tilde{x}_2' | x_1). \tag{43}$$

Proof. By a slight abuse of notation, we use interchangeably the distribution $\tilde{x}$ and a random variable $\tilde{x}$ that follows this distribution. Denote alose $\mathbf{S}_1 \triangleq \text{Supp}(\tilde{x}_1) \cup \text{Supp}(\tilde{x}_1')$. Our proof essentially boils down to the triangle inequality. More precisely, we have

$$\begin{aligned}
\text{TVDist}(\tilde{x}, \tilde{x}') &= \frac{1}{2} \sum_{x \in \mathbb{X}_1 \times \mathbb{X}_2} |\mathbb{P}[\tilde{x} = x] - \mathbb{P}[\tilde{x}' = x]| && (44)\\
&= \frac{1}{2} \sum_{(x_1, x_2) \in \mathbb{X}_1 \times \mathbb{X}_2} |\mathbb{P}[\tilde{x}_1 = x_1 \wedge \tilde{x}_2 = x_2] - \mathbb{P}[\tilde{x}_1' = x_1 \wedge \tilde{x}_2' = x_2]| && (45)\\
&= \frac{1}{2} \sum_{(x_1, x_2) \in \mathbb{X}_1 \times \mathbb{X}_2} |\mathbb{P}[\tilde{x}_1 = x_1] \mathbb{P}[\tilde{x}_2 = x_2 \mid \tilde{x}_1 = x_1] - \mathbb{P}[\tilde{x}_1' = x_1] \mathbb{P}[\tilde{x}_2' = x_2 \mid \tilde{x}_1' = x_1]| && (46)\\
&\leq \frac{1}{2} \sum_{(x_1, x_2) \in \mathbb{X}_1 \times \mathbb{X}_2} |\mathbb{P}[\tilde{x}_1 = x_1] \mathbb{P}[\tilde{x}_2 = x_2 \mid \tilde{x}_1 = x_1] - \mathbb{P}[\tilde{x}_1' = x_1] \mathbb{P}[\tilde{x}_2 = x_2 \mid \tilde{x}_1 = x_1]| \\
&\quad + \frac{1}{2} \sum_{(x_1, x_2) \in \mathbb{X}_1 \times \mathbb{X}_2} |\mathbb{P}[\tilde{x}_1' = x_1] \mathbb{P}[\tilde{x}_2 = x_2 \mid \tilde{x}_1 = x_1] - \mathbb{P}[\tilde{x}_1' = x_1] \mathbb{P}[\tilde{x}_2' = x_2 \mid \tilde{x}_1' = x_1]| && (47)\\
&\leq \frac{1}{2} \sum_{x_1 \in \mathbb{X}_1} |\mathbb{P}[\tilde{x}_1 = x_1] - \mathbb{P}[\tilde{x}_1' = x_1]| \underbrace{\sum_{x_2 \in \mathbb{X}_2} \mathbb{P}[\tilde{x}_2 = x_2 \mid \tilde{x}_1 = x_1]}_{=1} \\
&\quad + \sum_{x_1 \in \mathbb{X}_1} \mathbb{P}[\tilde{x}_1' = x_1] \cdot \underbrace{\frac{1}{2} \sum_{x_2 \in \mathbb{X}_2} |\mathbb{P}[\tilde{x}_2 = x_2 \mid \tilde{x}_1 = x_1] - \mathbb{P}[\tilde{x}_2' = x_2 \mid \tilde{x}_1' = x_1]|}_{=\text{TVDist}(\tilde{x}_2 | x_1, \tilde{x}_2' | x_1)} && (48)\\
&\leq \frac{1}{2} \sum_{x_1 \in \mathbb{X}_1} |\mathbb{P}[\tilde{x}_1 = x_1] - \mathbb{P}[\tilde{x}_1' = x_1]| + \left(\sup_{x_1 \in \mathbb{X}_1} \text{TVDist}(\tilde{x}_2 | x_1, \tilde{x}_2' | x_1) \right) \underbrace{\sum_{x_1 \in \mathbf{S}_1} \mathbb{P}[\tilde{x}_1' = x_1]}_{=1} && (49)\\
&= \text{TVDist}(\tilde{x}_1, \tilde{x}_1') + \sup_{x_1 \in \mathbf{S}_1} \text{TVDist}(\tilde{x}_2 | x_1, \tilde{x}_2' | x_1), && (50)
\end{aligned}$$

which is the lemma. □

B.3 Proof of Theorem 3

Proof of Theorem 3. Our proof goes by induction. The case $k = 1$ is a direct result of Lemma 1 and Lemma 2 (using the fact that $\|\cdot\|_{1,1} = \|\cdot\|_1$).

Now assume that

$$\text{TVDist}\left(\text{VECHE0}_{1:k}(v,b), \text{VECHE0}_{1:k}(v',b),\right) \leq \|v-v'\|_1 \sum_{i=1}^{k} \|W(v,b)\|_{1,i}^{-1}. \tag{51}$$

We then note that $\text{VECHE0}_{1:k+1}$ can be viewed as a distribution over $E^k \times E$, i.e. it first selects the first k entities of the feed, and then the $(k+1)$-th entity. Denote $\mathcal{F}_k^E \triangleq \left\{ f \in [E]^k \,\middle|\, |f([k])| = k \right\}$ the set of feeds of k all-different entities and $d \triangleq \text{TVDist}\left(\text{VECHE0}_{1:k+1}(v,b), \text{VECHE0}_{1:k+1}(v',b),\right)$ the distance we want to upper-bound. By Lemma 3, we have

$$d \leq \|v-v'\|_1 \sum_{i=1}^{k} \|W(v,b)\|_{1,i}^{-1}. + \sup_{f_{1:k} \in \mathcal{F}_k^E} \text{TVDist}\left(\text{VECHE0}_{k+1}(v,b)|f_{1:k}, \text{VECHE0}_{k+1}(v',b)|f_{1:k},\right). \tag{52}$$

We now remark that $k+1$-th entity to be drawn is drawn as described in Lemma 2, but with the vectors $\mathrm{W}(v,b)_{|[E]-f_{1:k}([k])}$ and $\mathrm{W}(v',b)_{|[E]-f_{1:k}([k])}$, i.e. with the weight vectors where the weights of the previously drawn entities $e \in f_{1:k}([k])$ are nullified. Therefore,

$$d \leq \|v-v'\|_1 \sum_{i=1}^{k} \|W(v,b)\|_{1,i}^{-1}. + \sup_{f_{1:k} \in \mathcal{F}_k^E} \frac{\left\|\mathrm{W}(v,b)_{|[E]-f_{1:k}([k])} - \mathrm{W}(v',b)_{|[E]-f_{1:k}([k])}\right\|_1}{\left\|\mathrm{W}(v,b)_{|[E]-f_{1:k}([k])}\right\|_1}. \tag{53}$$

We now observe that canceling some coordinates can only reduce the ℓ_1 distance between two vectors, thus $\left\|\mathrm{W}(v,b)_{|[E]-f_{1:k}([k])} - \mathrm{W}(v',b)_{|[E]-f_{1:k}([k])}\right\|_1 \leq \|\mathrm{W}(v,b) - \mathrm{W}(v',b)\|_1$. By Lemma 1, we know that this is at most $\|v-v'\|_1$. Hence

$$d \leq \|v-v'\|_1 \sum_{i=1}^{k} \|W(v,b)\|_{1,i}^{-1}. + \|v-v'\|_1 \sup_{f_{1:k} \in \mathcal{F}_k^E} \left\|\mathrm{W}(v,b)_{|[E]-f_{1:k}([k])}\right\|_1^{-1} \tag{54}$$

We now note that

$$\sup_{f_{1:k} \in \mathcal{F}_k^E} \left\|\mathrm{W}(v,b)_{|[E]-f_{1:k}([k])}\right\|_1^{-1} \leq \left(\inf_{\substack{\mathbf{F} \subset [E] \\ |\mathbf{F}|=k}} \left\|\mathrm{W}(v,b)_{|[E]-\mathbf{F}}\right\|_1 \right)^{-1} \leq \|\mathrm{W}(v,b)\|_{1,k+1}^{-1}. \tag{55}$$

Using additionally that we trivially have $\|W(v,b)\|_{1,k} \geq \|W(v,b)\|_{1,k+1}$, we finally obtain

$$d \leq \|v-v'\|_1 \sum_{i=1}^{k} \|W(v,b)\|_{1,i}^{-1}. + \frac{\|v-v'\|_1}{\|\mathrm{W}(v,b)\|_{1,k+1}} = \|v-v'\|_1 \sum_{i=1}^{k+1} \|W(v,b)\|_{1,i}^{-1}, \tag{56}$$

which concludes the proof. □

C Proofs of time decay bias properties

Proof of Proposition 2.

Proof. This is a straightforward computation. □

Proof of Proposition 3.

Proof. Denote $\varepsilon \triangleq (t_{uf} - t_{ue})/(t - t_{uf})$, $\eta \triangleq \tau/(t - t_{uf})$ and $R \triangleq \frac{b_{ue} \ln(1+\tau_{ue}/\tau_0)^{\gamma_\tau}}{b_{uf} \ln(1+\tau_{uf}/\tau_0)^{\gamma_\tau}}$. We have

$$\frac{1}{R}\frac{\textsc{Bias}_e^{time}(b_u)}{\textsc{Bias}_f^{time}(b_u)} = \frac{\tau^2 + (t - t_{ue})^2}{\tau^2 + (t - t_{uf})^2} = 1 + \frac{(t - t_{uf} + t_{uf} - t_{ue})^2 - (t - t_{uf})^2}{\tau^2 + (t - t_{uf})^2} \tag{57}$$

$$= 1 + \frac{(1+\varepsilon)^2 - 1}{\eta^2 + 1} = 1 + 2\varepsilon\left(1 + \frac{\varepsilon}{2}\right)\left(1 + \eta^2\right)^{-1} \tag{58}$$

$$= 1 + 2\varepsilon + O(\varepsilon^2) + O(\varepsilon\eta^2), \tag{59}$$

which concludes the proof. □

Proof of Proposition 4. The proof leverages a classical result.

Lemma 4. $\int_0^\infty (1 + x^2)^{-1} dx = \pi/2$.

Proof. This is a classical integral, which can be performed using the change of variable $x = \tan\theta$. □

Proof of Proposition 4. Assume without loss of generality that user u started acting at time 0, Fix $t \in \mathbb{R}_{>0}$ and denote I a number of time subdivisions. Denote B_i^I the number of actions taken in interval $\left[\frac{i}{I}t, \frac{i+1}{I}t\right]$. Given that the actions follow a Poisson point process, B_i follows a Poisson distribution of parameter $\lambda t/I$. Denote $b_u(t) \triangleq \textsc{Bias}(b_u|t, t_u, \tau_u, \tau_0, \gamma_\tau)$. Since

$$\|b_u(t)\|_1 \leq \sum_{i=0}^{I-1} \frac{B_i \tau^2}{\tau^2 + i^2 t^2/I^2}, \tag{60}$$

it follows that

$$\mathbb{E}\,\|b_u(t)\|_1 \leq \sum_{i=0}^{I-1} \frac{\lambda t}{I} \frac{\tau^2}{\tau^2 + i^2 t^2/I^2}. \tag{61}$$

Taking the limit $I \to \infty$ yields

$$\mathbb{E}\,\|b_u(t)\|_1 \leq \lambda \int_0^t \frac{\tau^2 ds}{\tau^2 + s^2} \leq \lambda \int_0^\infty \frac{ds}{1 + (s/\tau)^2} \overset{r=s/\tau}{=} \lambda \int_0^\infty \frac{\tau dr}{1 + r^2} = \lambda\pi/2,$$

using Lemma 4. This concludes the proof. □

Proof of Proposition 5.

Proof. Indeed we have

$$\int_{t_{ue}}^\infty \textsc{Bias}_e^{time}(b_u|t, t_{ue}, \tau_{ue}, \tau_0, \gamma_\tau) dt = \int_0^\infty b_{ue} \cdot \frac{\tau_{ue} \ln(1 + \tau_{ue}/\tau_0)^{\gamma_\tau}}{(t - t_{ue})^2 + \tau_{ue}^2} dt \tag{62}$$

$$= \frac{b_{ue}}{\tau_{ue}} \ln(1 + \tau_{ue}/\tau_0)^{\gamma_\tau} \int_0^\infty \frac{dt}{1 + \left(\frac{t - t_{ue}}{\tau_{ue}}\right)^2} \tag{63}$$

$$\overset{r=(t-t_{ue}/\tau_{ue})}{=} \frac{b_{ue}}{\tau_{ue}} \ln(1 + \tau_{ue}/\tau_0)^{\gamma_\tau} \int_0^\infty \frac{\tau_{ue} dr}{1 + r^2} = b_{ue} \cdot \frac{\pi}{2} \cdot \ln(1 + \tau_{ue}/\tau_0)^{\gamma_\tau}, \tag{64}$$

using Lemma 4. This concludes the proof. □

D Proof of guarantees using Tillé's pivot (Theorem 2)

Lemma 5. *The top recommendation of Tillé's algorithm matches that of* VECHE0*, i.e.* $\text{TILLEVECHE}_1(b) = \text{VECHE0}_1(b)$ *for all ballots* $b \in \mathbb{R}^{U \times E}$.

Proof. We prove this by induction over the number E of entities. If $E = 2$, then it is trivial.

Assume it holds for E entities. Consider $E + 1$ entities, and let $b \in \mathbb{R}^{U \times E}$ and denote $w = \mathrm{W}(b)$. Let $e^* = \text{TILLEVECHE}_1(b)$ be the random top entity, Let $(e_1, e_2) \in [E + 1]^2$ be the random first pair of entities to be opposed in the first round, with $e_1 \neq e_2$, and $x \in \{1, 2\}$ be the result of their opposition (with $x = 1$ if e_1 wins). By the law of total probability, we have

$$\mathbb{P}\left[e^* = e|b\right] = \mathbb{E}_{e_1,e_2}\left[\mathbb{E}_x\left[\mathbb{P}\left[e^* = e|w, e_1, e_2, x\right]\right]\right] \tag{65}$$

$$= \mathbb{E}_{e_1,e_2}\left[\frac{w_{e_1}\mathbb{P}\left[e^* = e|w^{e_1>e_2}\right] + w_{e_2}\mathbb{P}\left[e^* = e|w^{e_2>e_1}\right]}{w_{e_1} + w_{e_2}}\right], \tag{66}$$

where $w^{e_1>e_2}$ is the resulting weight vector after e_1 stealing the weights of e_2. By the induction assumption, the distribution of e^* given $w^{e_1>e_2}$ matches sampling without replacement, which leads to

$$\mathbb{P}\left[e^* = e|w^{e_1>e_2}\right] = \frac{w_e^{e_1>e_2}}{\|w^{e_1>e_2}\|_1} = \frac{w_e^{e_1>e_2}}{\|w\|_1}, \tag{67}$$

using the fact that the weights have been transfered but not "lost". For all $e \notin \{e_1, e_2\}$, we then have $\mathbb{P}\left[e^* = e|w^{e_1>e_2}\right] = w_e / \|w\|_1$. Moreover, $\mathbb{P}\left[e^* = e_1|w^{e_1>e_2}\right] = (w_{e_1} + w_{e_2}) / \|w\|_1$ and $\mathbb{P}\left[e^* = e_2|w^{e_1>e_2}\right] = 0$. Overall, for all $e \in [E + 1]$, we thus have

$$\frac{w_{e_1}\mathbb{P}\left[e^* = e|w^{e_1>e_2}\right] + w_{e_2}\mathbb{P}\left[e^* = e|w^{e_2>e_1}\right]}{w_{e_1} + w_{e_2}} = \frac{w_e}{\|w\|_1}. \tag{68}$$

Plugging this into the total law equation (66) yields the lemma. □

Proof of Theorem 2. This follows straightforwardly from Theorem 1 and Lemma 5. □

E Geometric median as an aggregation rule

The way we aggregated users' votes was a simple averaging. While this has the great advantage of being a simple mechanism, and while it already provides great guarantees, it could be argued to be incompletely satisfactory. For instance, one issue is that a user may view an already widely reposted entity as worth recommending, but they may not want to "waste" their vote to promote it; they may prefer to promote another insufficiently reposted entity.

Perhaps a more satisfactory aggregation rule could be the geometric median. Namely, denote $\mathbf{E}_u \subset [E]$ the subset of entities that user u posted, reposted, evaluated or reported. Let $c_u \triangleq (b_{ue}^+)_{e \in \mathbf{E}_u} \in \mathbb{R}_{\geq 0}^{\mathbf{E}_u}$ the nonnegative part of the ballots for entities on which an action was taken. Then $\tilde{c}_u \triangleq c_u / \|c_u\|_1$ its ℓ_1 normalization, which can be viewed as a preferred probability distribution, if only one evaluated entity had to be recommended. The geometric median would consist of selecting a weight vector that minimizes its distance to the user's preferred distribution (here, we also ignore non-evaluated entities), yielding

$$w \triangleq \underset{\omega \in \mathbb{R}^E}{\arg\min} \sum_{u \in [U]} v_u \left\|P_{\mathbf{E}_u}\omega - \tilde{c}_u\right\|_2, \tag{69}$$

where $P_{\mathbf{E}_u}$ is the orthogonal projection on $\mathbf{E}_u$. This corresponds to the geometric median with skewed preferences defined in [EFGH23].